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Chemical reactions and adherence at glass/metal interfaces: an analysis

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Abstract

A review of wetting, spreading and reactions of sodium disilicate glass on metals is presented. The selected metals represent a wide range of reactivity. Most of them are either principal constituents of dental alloys or are present as alloying additions. Conditions for the occurrence of the different types of redox reactions at glass/metal interfaces are analyzed. All of the reactions have an effect in some way, both chemically and physically, on the bonding of glass to metal.

Key words

Outline

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glass/metal interface reactions; wetting and spreading; adherence

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